

Geotextile Reinforced Ramp

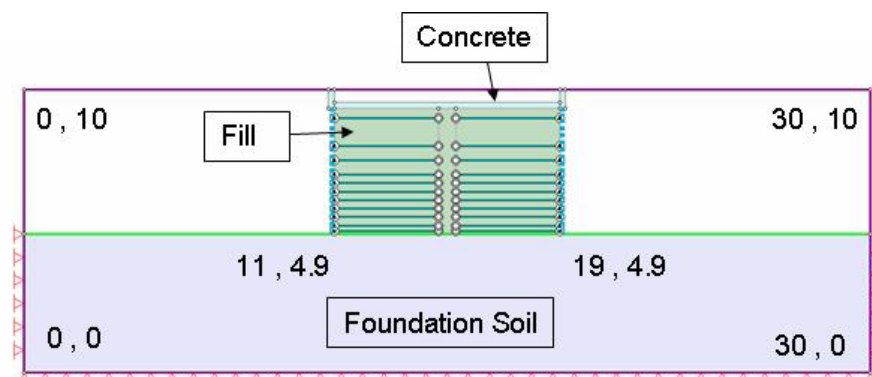
In this tutorial, a ramp is constructed and its performance under loading is assessed. The model is created in four stages as follows:

1. Foundation soil layer is brought to equilibrium.
2. Fill is added on top of the soil interlayed with geotextile support layers. Precast concrete liners are added to support the fill.
3. The concrete road bed is constructed on top of the fill.
4. Load is applied to the road surface.

Topics Covered

- Import DXF
- Geotextiles
- Liners
- Structural Interfaces
- Selection Windows
- Selection Filter
- Discretization Density
- Mapped meshing
- Staged Loading
- Liner Bending Moments

Geometry

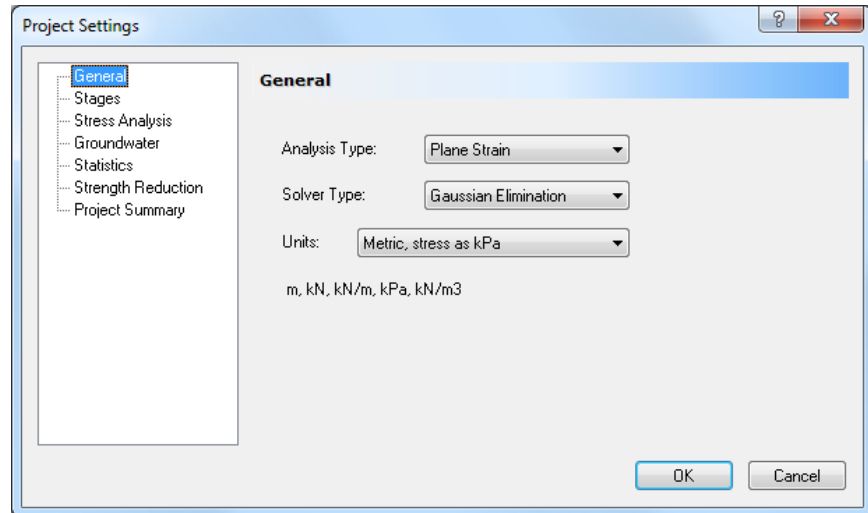


Model

Start the *Phase2* Model program.

Project Settings

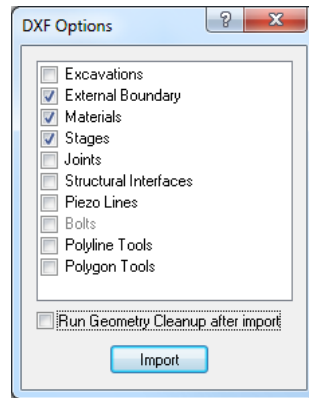
Open the Project Settings dialog from the Analysis menu and make sure the General tab is selected. Define the units as being “Metric, stress as kPa”. Change the number of stages to 4. Close the dialog by clicking OK.



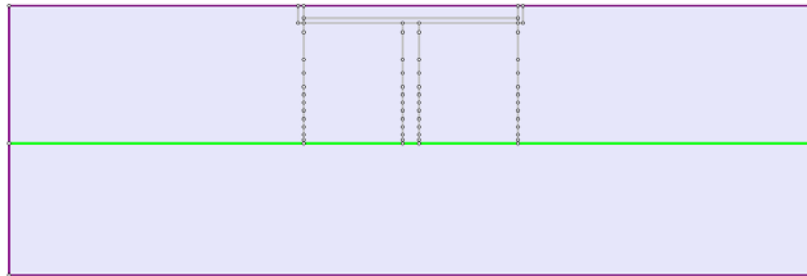
If you changed the Units from some other system, then you will see a warning window telling you that default values will change to reflect the new units. Click Yes to verify the unit change.

Boundaries

To save typing in many coordinates, we will import the boundary geometry from a DXF file (AutoCAD Drawing Exchange File). This technique is useful if a problem geometry has already been constructed in Autocad and you do not wish to re-enter the geometry in *Phase2*. To import the file, select **Import DXF** from the **Import** sub-menu of the **File** menu. In the DXF Options dialog, select External Boundary, Materials, and Stages and deselect Run Geometry Cleanup after import, as shown.



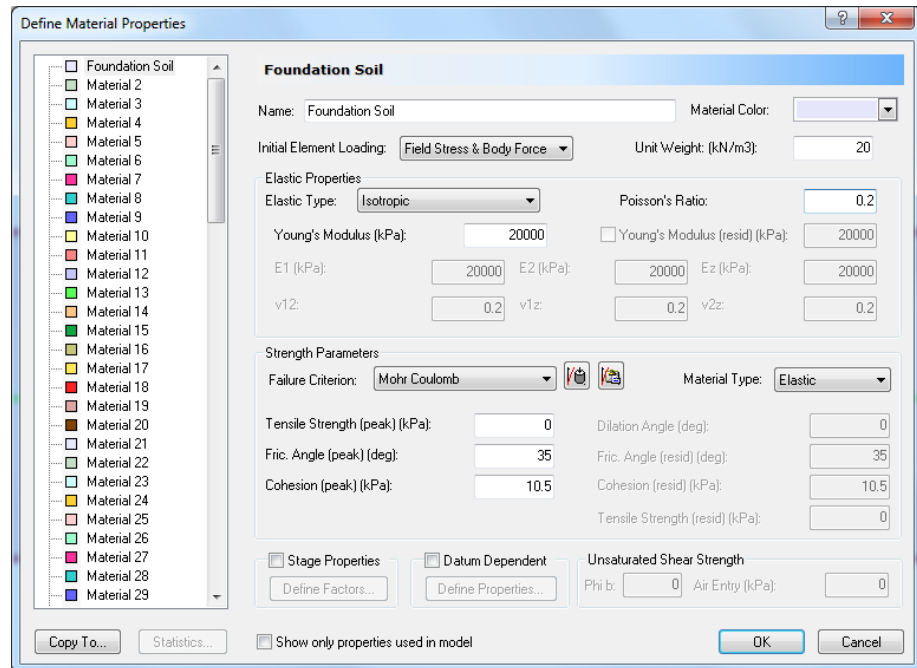
Select the Import button, and you will see an Open File dialog. Select the file **Tutorial 14 boundaries.dxf** from the Examples > Tutorials folder in your *Phase2* installation folder. Click OK. You will see a model with external, material and stage boundaries as shown.



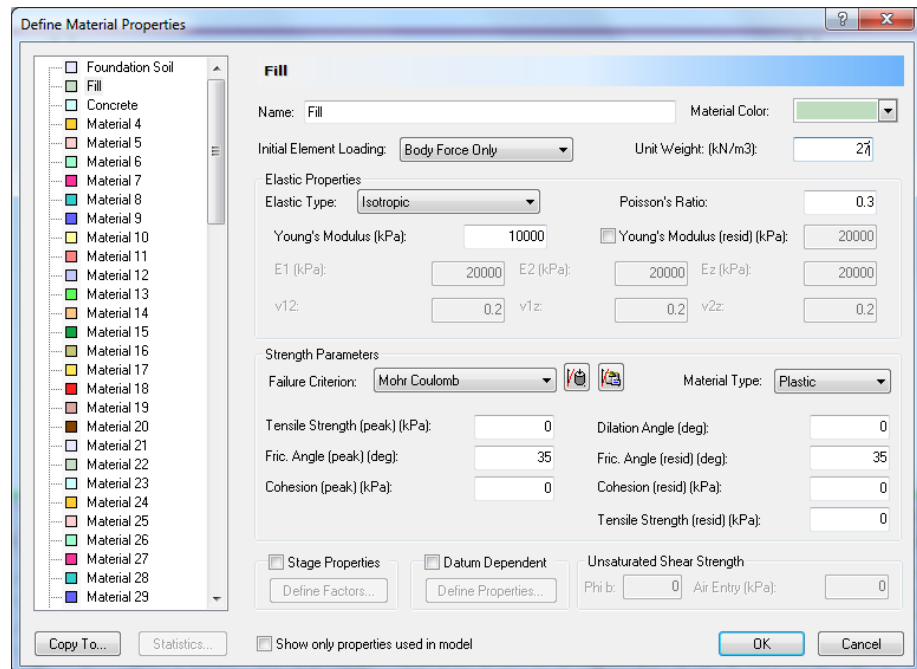
NOTE: Stage boundaries perform the same function as material boundaries but are plotted grey instead of green. In previous versions of *Phase2*, stage boundaries were used to delineate excavations between stages and material boundaries were used to separate different materials. You can now use them interchangeably.

Material Properties

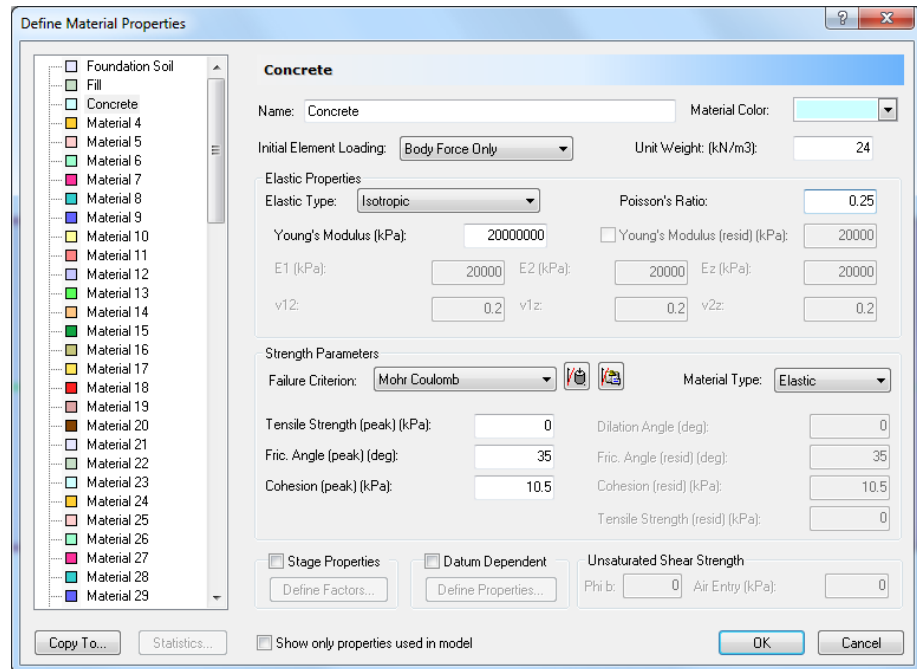
Select Define Materials from the Properties menu. Change the name of Material 1 to Foundation Soil. For Initial Element Loading choose Field Stress & Body Force. This choice reflects that the soil layer is in-situ with both self weight and stress. See the *Phase2* Help for more information on initial element loading. Set the Unit Weight to 20 kN/m³ and Poisson's Ratio to 0.2. Leave all other properties as the default values. Note that the Material Type is set to elastic, meaning that there can be no failure in this layer.



Now click on the Material 2 tab. Change the name of Material 2 to Fill. Change the Initial Element Loading to Body Force only. This is the logical choice if the fill is to be manually added such that it settles under its own weight. Change the Unit Weight to 27 kN/m³ and set the Young's Modulus to 10000 kPa. Under Strength Parameters, set the Material Type to plastic to enable failure in the Fill. Set the peak and residual friction angle to 35 degrees and the peak and residual cohesion to 0 kPa as shown.



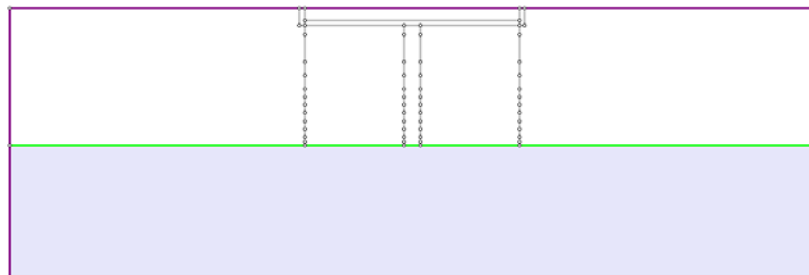
Now click on the tab for Material 3. Change the Name to Concrete and enter the parameters as shown below. Note the high value for Young's Modulus indicating a very stiff material. Observe also that the Material Type is elastic, indicating that there is no failure allowed in the Concrete.



Click OK to close the dialog.

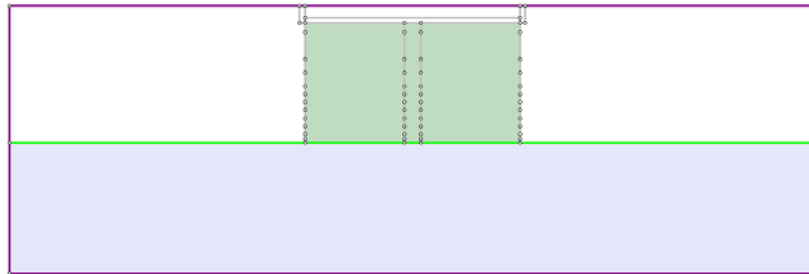
Assign Material Properties

By default, the entire model is initially assigned the properties of Foundation Soil (material 1). We now need to build up the road material stage by stage. Click on the Stage 1 tab at the bottom of the *Phase2* window. Choose Assign Properties from the Properties menu. Select excavate and click in all areas above the soil material boundary. This should leave just the Foundation Soil as shown.



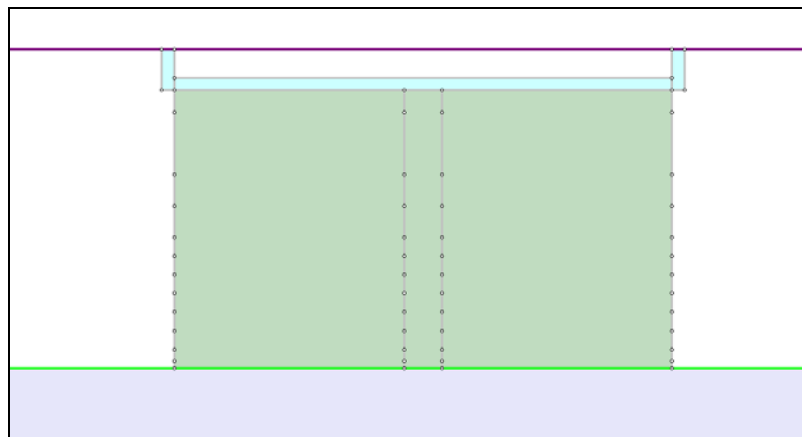
Assigned materials for Stage 1.

Now click the Stage 2 tab. Select Fill from the Assign Materials dialog and click inside the three sections that make up the fill as shown.



Assigned materials for Stage 2.

Click on the Stage 3 tab. Select Concrete from the Assign Material dialog and click inside the three sections making up the concrete road surface as shown.

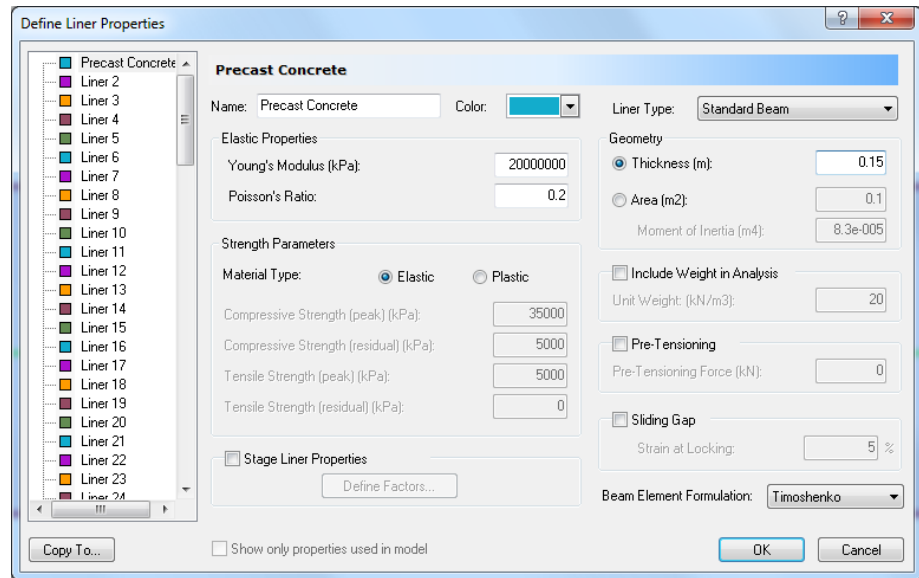


Assigned materials for Stage 3.

Close the Assign Materials dialog.

Support Properties

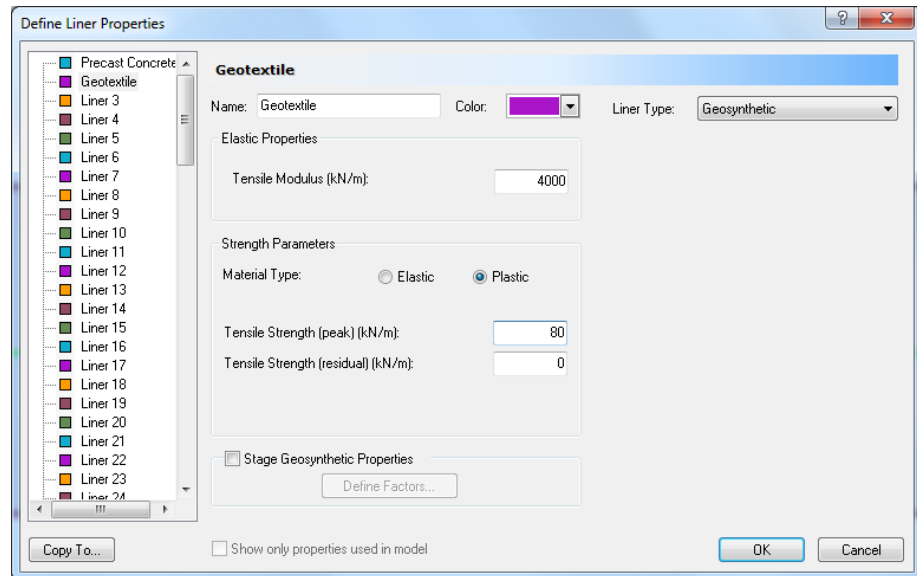
First we will set the properties of the precast concrete support. Choose Define Liners from the Properties menu. Change the name from Liner 1 to Precast Concrete. Change the Young's modulus to 20000000 kPa and the thickness to 0.15 m as shown.



Click OK to close the dialog.

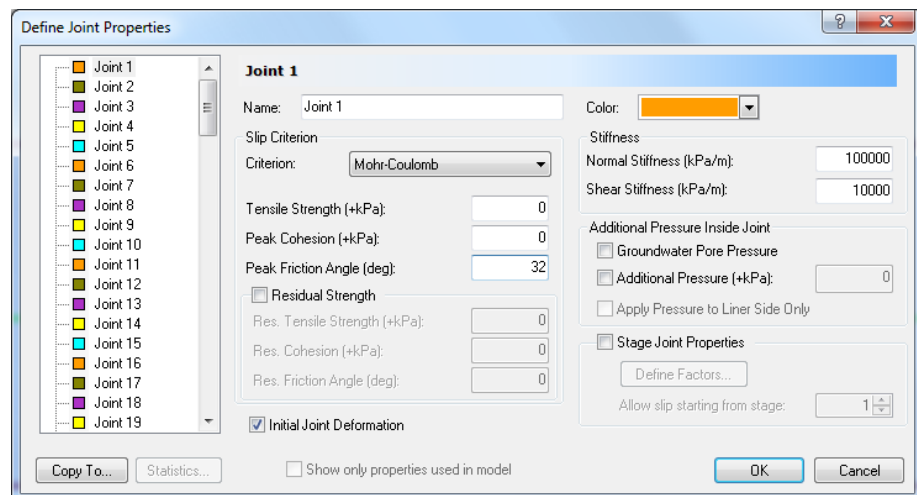
The geotextile reinforcement layers must be modelled using Structural Interfaces. These allow slip to occur between the fill material and the geotextile. To set the properties of the Structural Interface select Define Structural Interface from the Properties menu. You will now see how a structural interface is composed of a liner sandwiched between two joints (slip elements). We don't want to use the precast concrete as our geotextile liner, so for the Liner choose Liner 2 from the drop down menu.

To set the properties of the Liner, click on the ellipsis (...) to the right of Liner 2. You will now see the Define Liner Properties dialog with Liner 2 selected. Change the Name to Geotextile. Under Liner Type select Geosynthetic. Now set the tensile modulus to 4000 kN/m, the Material Type to Plastic and the peak Tensile strength to 80 kN/m as shown. Leave the residual tensile strength at 0.



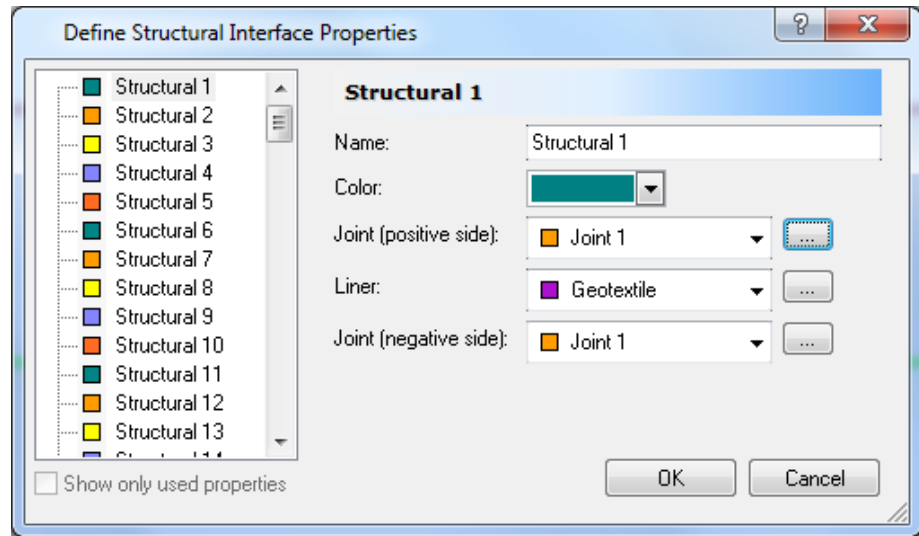
Click OK to close the dialog.

You now need to set the properties of the joints on either side of the liner. In the Define Structural Interface Properties dialog, click on the ellipsis (...) to the right of Joint 1. You will now see the Define Joint Properties dialog. Change the Slip Criterion to Mohr-Coulomb and set the Friction Angle to 32 degrees as shown.



Click OK to close the dialog.

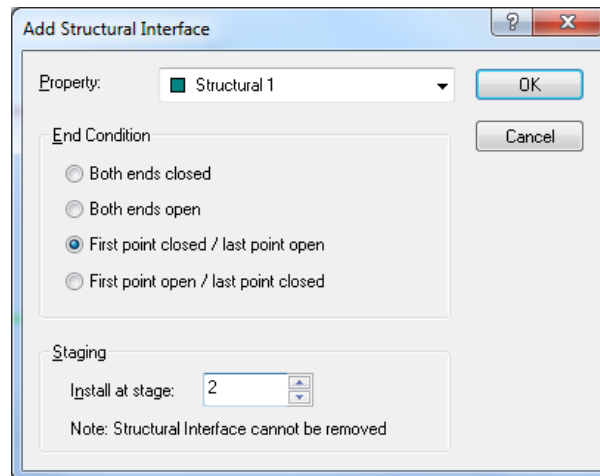
Your Structural Interface Properties dialog should now appear as shown.



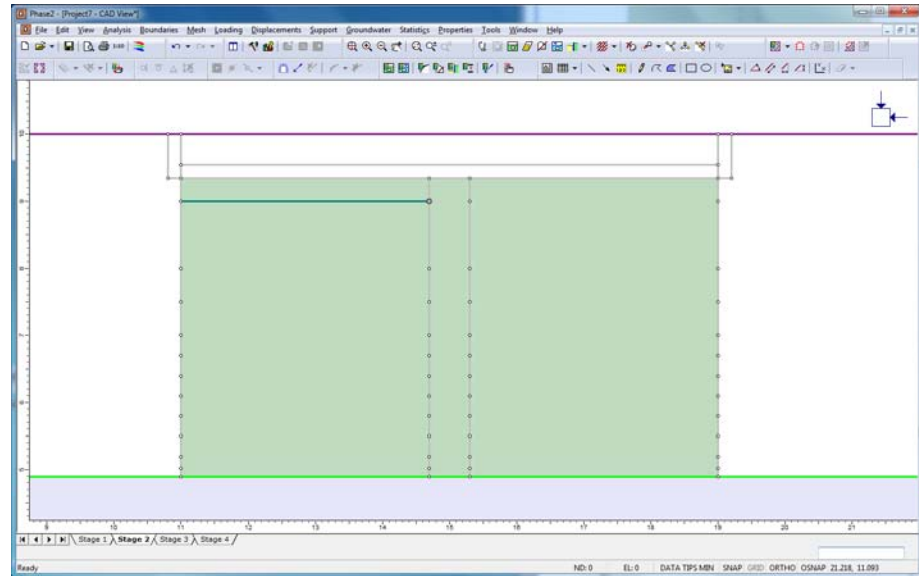
Click OK to close the dialog.

Assign Geotextiles

The geotextile supports are a series of horizontal Structural Interfaces within the Fill. The coordinates are already defined from reading in the DXF file. To add the Structural Interfaces, select Add Structural Interface from the Boundaries menu. In the Add Structural Interface dialog, select **First point closed / last point open**. Under Staging, set **Install at stage** to 2. The dialog should appear as shown.

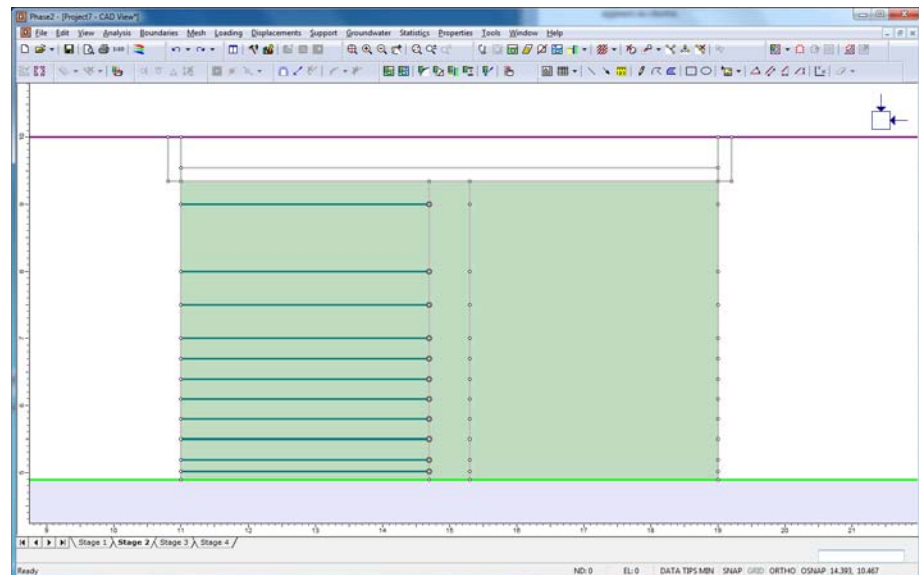


Click the OK button. Select the point with coordinates (11, 9). You may enter the point with the keyboard but it is easy to select the point with the mouse since the cursor will snap to existing points. Now click on the point at (14.7, 9). Right click and choose Done. You will now have one structural interface that is free to slip on the inside edge and is unable to slip on the outside edge.



TIP: to enable snapping to existing geometry, select Snap from the View menu and turn on all options. You can perform the same task by clicking on the words in the status bar at the bottom of the *Phase2* window next to the coordinates, or use the right-click menu.

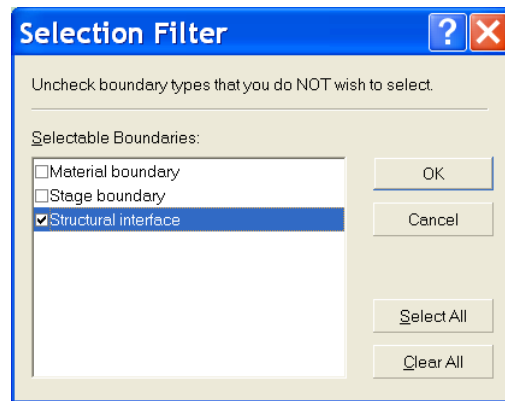
The easiest way to add the remaining Structural Interfaces is by copying and pasting. Right click on the Structural Interface you just constructed. Select Copy Boundary. You now need to enter a point from which to copy (base point), so select the left point of the existing Structural Interface. Enter the point to copy to by selecting the next point directly below (11, 8). A copy of the Structural Interface should now appear. Repeat these steps for the remaining points on the left side of the Fill until your model appears as shown.



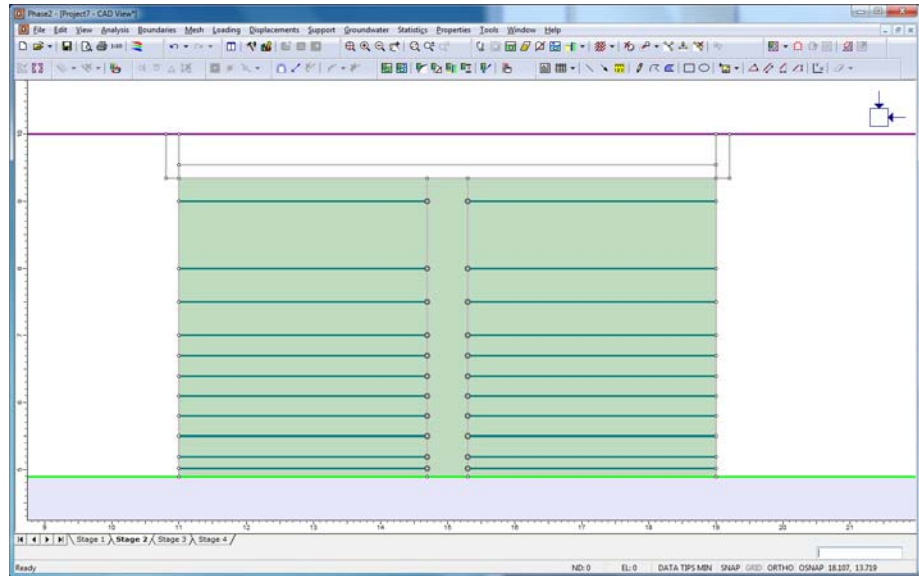
Now we need to add Structural Interfaces to the right side of the fill. You could repeat the above technique (be careful to ensure that the outside point is closed and the inside point is open). Alternatively you could use the multiple copying capability of *Phase2*. To do this, go to the Boundaries menu, select Edit and choose Copy. You can now select all of the Structural Interfaces by drawing a selection window around them (click the left mouse button and hold it down while moving the mouse to expand the size of the box). Hit Enter to finish the selection. Now select a base point to copy from (say 11 , 9) and then a point to copy to (15.3, 9). You should now have an exact copy of the Interfaces on the right side of the fill.

TIP 1: For selecting multiple boundaries, the default option is *Select Inside Only*, meaning that only line segments completely inside the selection window will be selected. You can change this to select line segments that cross the selection window as well by right clicking while you are making your selection, choosing Selection Window, and choosing Select Inside or Crossing.

TIP 2: When trying to select the Structural Interfaces, you may enclose other boundaries with your Selection Window. Since we only want to copy the Structural Interfaces, we can use a Selection Filter. Just before you draw your Selection Window, right click and choose Selection Filter. Now uncheck everything except for Structural Interfaces as shown. This will ensure that only Structural Interfaces are selected.



Notice that the Interfaces you have just copied are oriented the wrong way, with the closed points on the inside and the open points on the outside. To flip them around, go to Boundaries, select Edit and choose Convert Boundary. Select all of the Interfaces on the right side using a Selection Window (don't forget to apply the Selection Filter again to ensure only Structural Interfaces are selected). You will now see the Convert To dialog box. We do not wish to change the type of boundary so select Structural Interface and click OK. In the Add Structural Interface dialog, select **First point open / last point closed**. Click OK. Your model should now look like this.

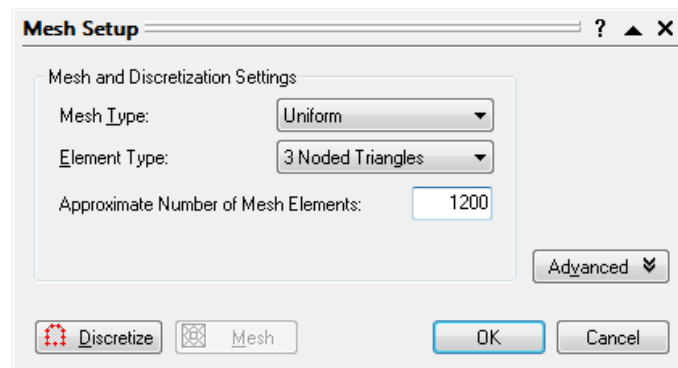


Finally we need to add the precast concrete liner. However you cannot add Liners until the finite element mesh exists. The next step is therefore to build the mesh.

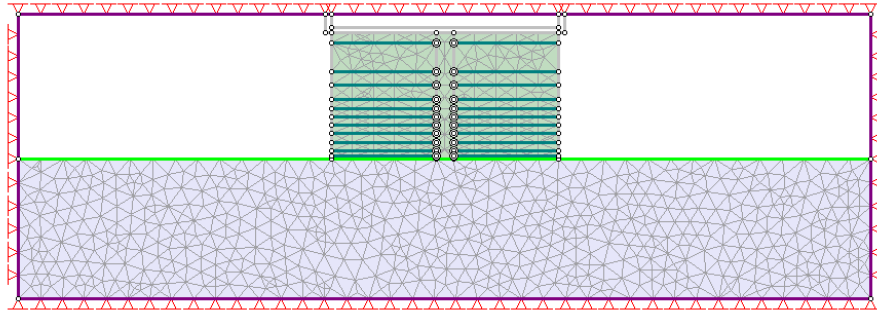
NOTE: Structural interfaces must be added *before* the mesh since the joints will affect the mesh generation. See the *Phase2* Help for more information.

Mesh

Select the **Mesh Setup** option in the **Mesh** menu. Change the Mesh Type to Uniform. Leave the default element type (3 Noded Triangles) but set the number of elements to 1200. Click the Discretize button followed by the Mesh button.

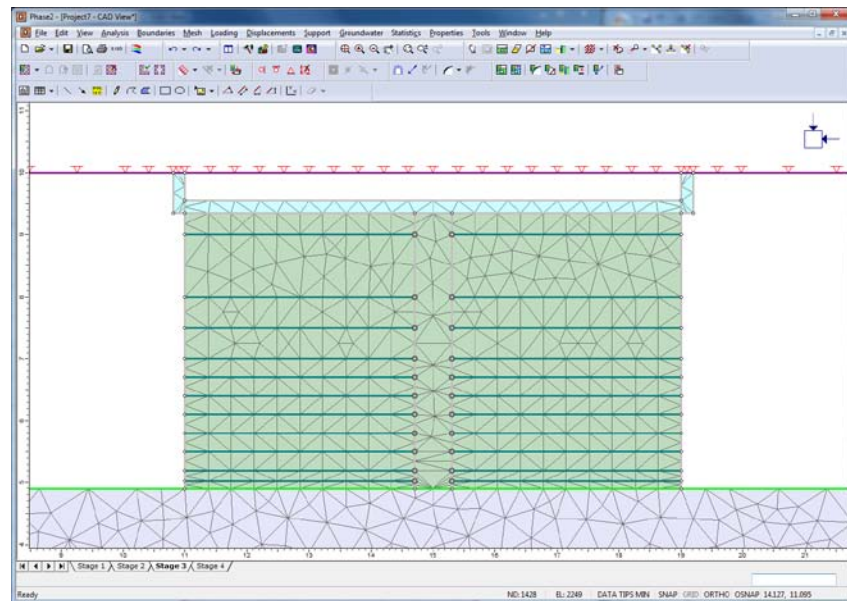


Close the Mesh Setup dialog by selecting the OK button. Your model should now appear as shown.



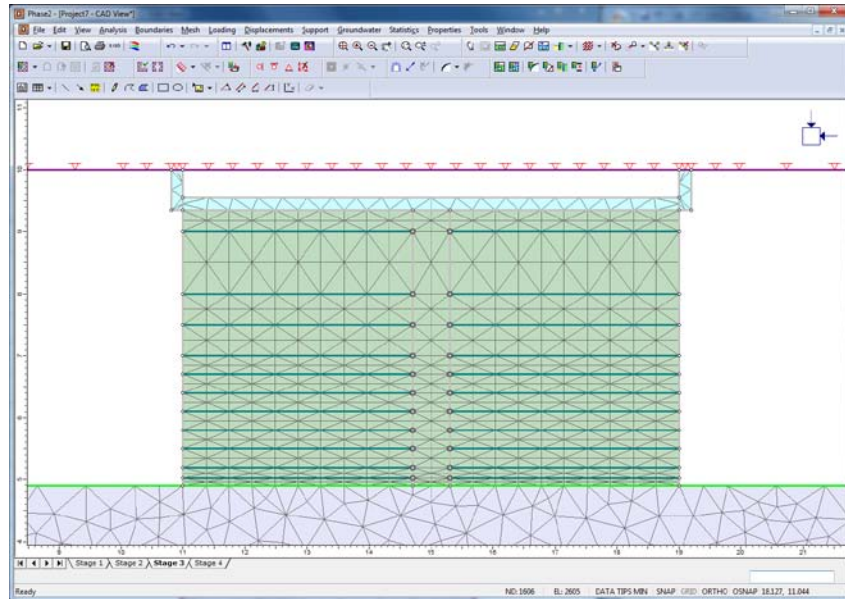
Initial mesh

We would like to improve the quality of the mesh in the region of interest (the ramp construction). To increase the density of the mesh in this area, first make sure you are looking at Stage 3 or Stage 4. Select Increase Discretization Density (window) from the Mesh menu. Draw a box around the road construction. You will now see a higher density mesh in the Fill and Concrete layers as shown.



Mesh after increasing discretization density in the ramp construction

To improve the mesh further, we can use the Mapped Meshing capability in *Phase2*. This generates a geometrically regular mesh in triangular or quadrilateral regions of the model, sometimes called a Structured Mesh. See the *Phase2* Help for more details. From the Mesh menu, select Mapped Meshing and then Automatic Mapped Mesh. You will see a dialog box with information about the Mapped Meshing. Click OK and your model should now look like this.

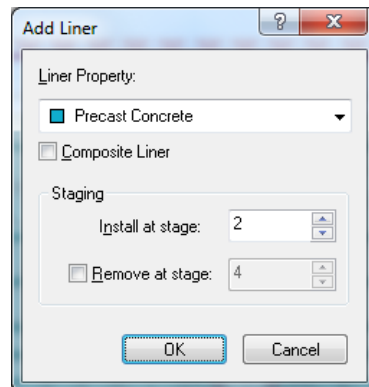


Mesh after mapped meshing

We have finished generating the finite element mesh so we can now add the precast concrete liners.

Assign Liners

Click on the Stage 2 tab. From the Support menu select Add Liner. Ensure that the Liner property is Precast Concrete and the **Install at stage** is set to 2.

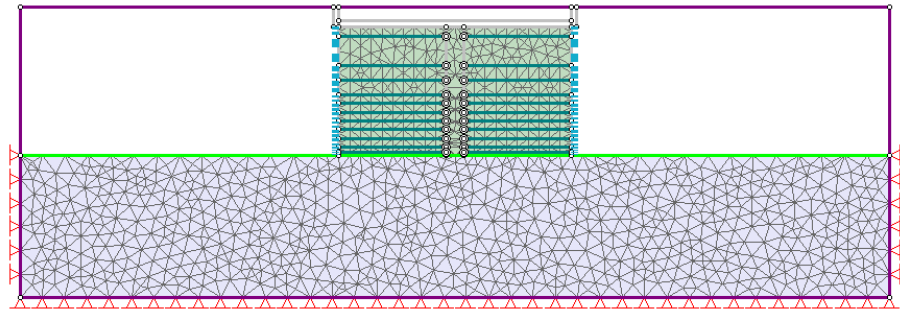


Click OK. Now select all of the boundary segments on the left side of the Fill layer. You can select them all at once by encompassing them in a box as described above. Do the same for the right side of the fill. Right click and choose Done Selection. Your model should now appear as shown.

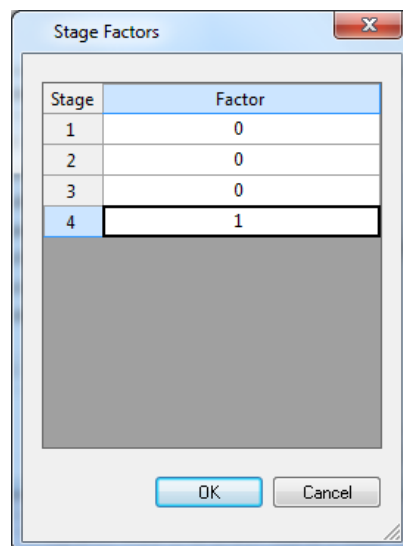
Boundary Conditions

The model now shows the default boundary conditions (all external boundaries fixed in the x and y directions). In our model, only the edges and bottom of the Soil layer need to be fixed. To free the other boundaries, select Free from the Displacements menu. Click on all of the outer boundaries except the sides and edges of the Soil layer. Be sure to select the very small boundary segments that form the top edges of the road (you may need to zoom in to select these). Hit Enter to finish selecting.

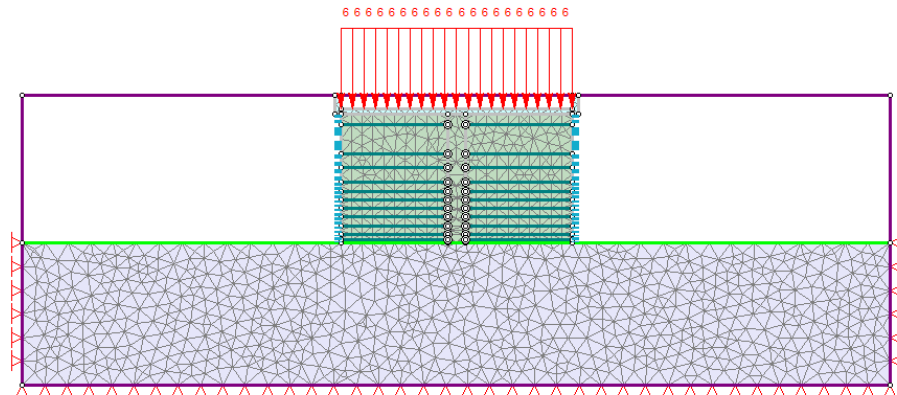
The top left and top right nodes of the soil layer have now been freed. To re-fix them, select Restrain X,Y from the Displacements menu and click on the left and right edges of the soil layer. Hit Enter to finish. Your model should now look like this.



We now need to add the loading to the road surface. Click on the Stage 4 tab. Under the Loading menu choose Distributed Loads and then Add Uniform Load. Leave the orientation as **Normal to boundary** and enter 6 kN/m for the magnitude. We only wish to apply the load during Stage 4, so turn on the Stage Load option and click the Stage Factors button. Enter 0 for the Stage 1, 2 and 3 Factors as shown.



Click OK. Now click OK in the Add Distributed Load dialog. Click on the stage boundary that is the road surface. Hit Enter and your model should now look like this.

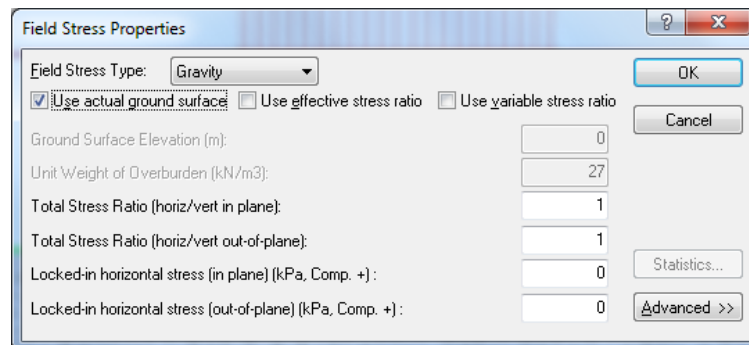


Boundary conditions for Stage 4.

You can click through the other stages to ensure that load is only being applied in Stage 4.

Field Stress

The last step is to specify the field stress in the model. Select Field Stress from the Loading menu. Set the Field Stress Type to Gravity and turn on the option **Use actual ground surface** as shown.



Click OK to close the dialog.

You have completed the definition of the model. Save the model using the Save option in the File menu.

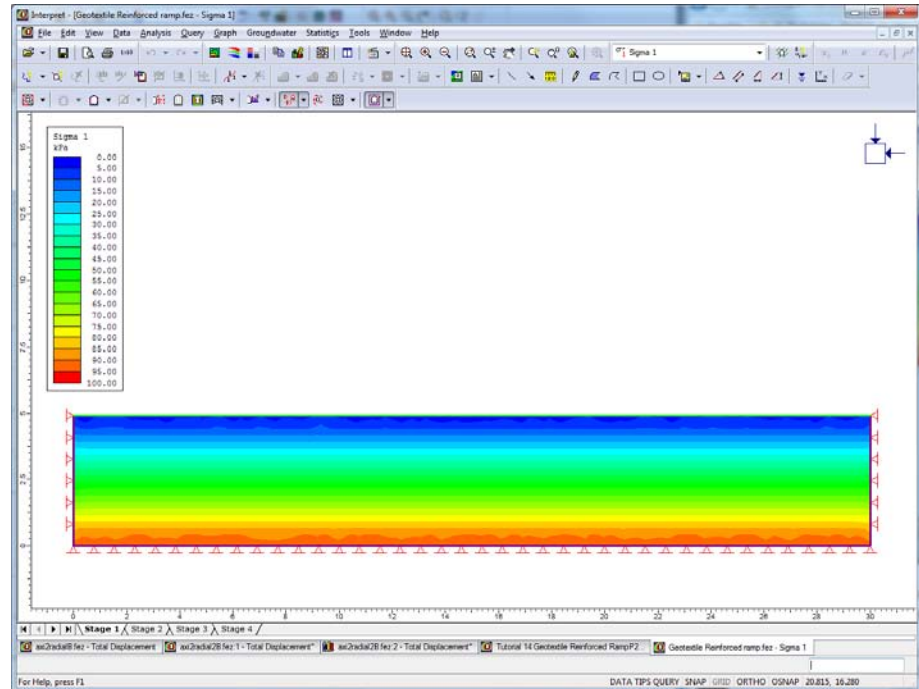
Compute

Run the model using the Compute option in the Analysis menu. The analysis should take under a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

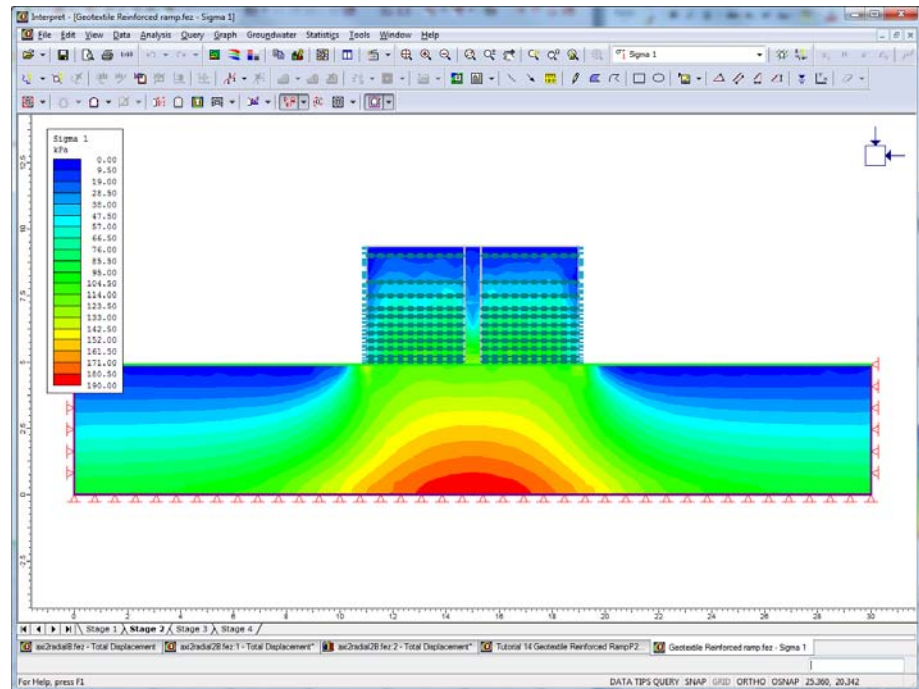
Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. The following screen is displayed showing the maximum stress in the Foundation Soil layer for Stage 1.

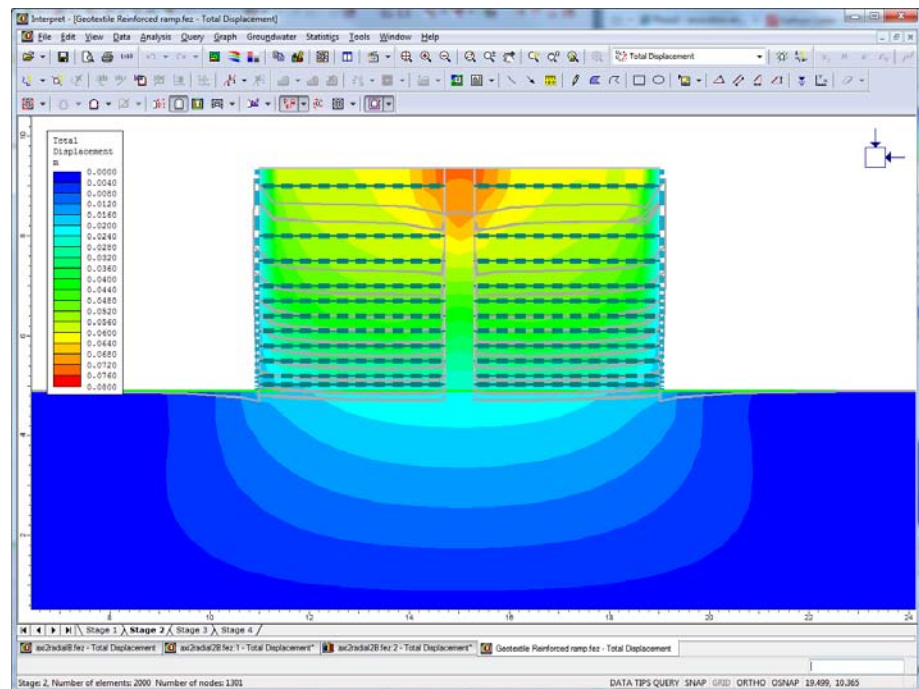


This reflects the initial element loading of the soil layer. If you plot displacements you will see no displacement has occurred.

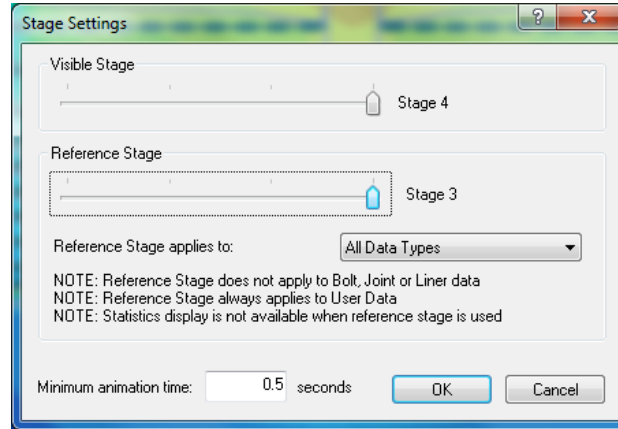
Click on the Stage 2 tab. You will now see the maximum stress after the fill has been added. To make the plot clearer you may want to shrink the size of the plotted Structural Interfaces. Right click anywhere in the window and choose Display Options. Click on the Stress tab and turn off the option for Expand Composites under the Support heading. Your plot should now look like this.



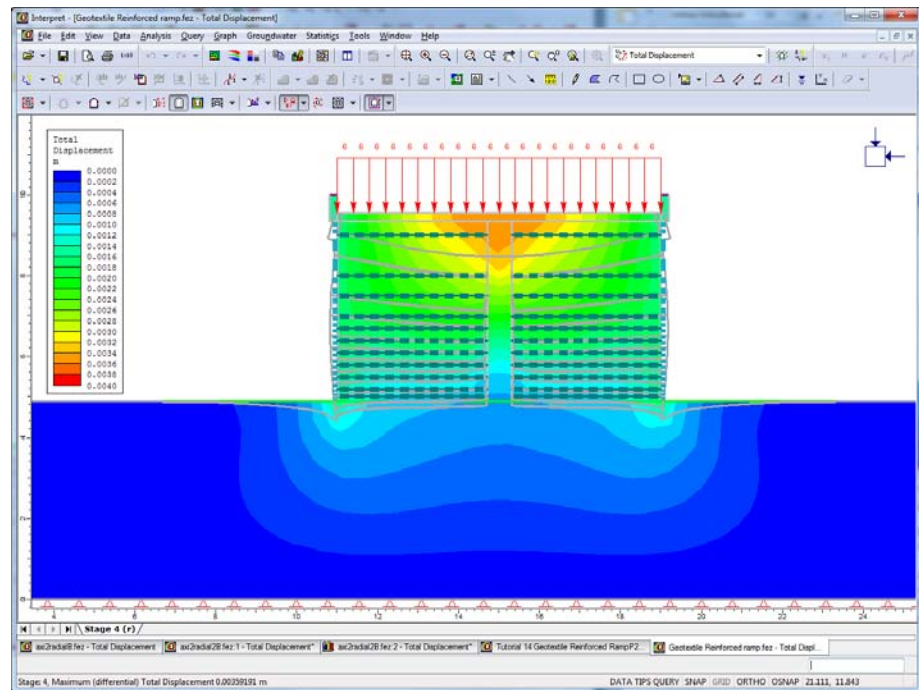
Plot displacements by choosing Total Displacement from the drop down menu on the toolbar. Plot the deformed boundaries by clicking the Display Deformed Boundaries button. You will see significant downward displacement in the fill since it has settled under its own weight. There is less displacement at the edges where the fill is being supported by the precast concrete liner.



Click through the other stages and you will see increased deformation as the concrete road bed is added and then when the load is applied in Stage 4. If you wish to see the effect of only the load, you can set Stage 3 as the reference stage. Do this by going to the Data menu, selecting Stage Settings and pushing the slider for Reference Stage to Stage 3.



Click OK. You will now see the effect of just the loading in the final stage.



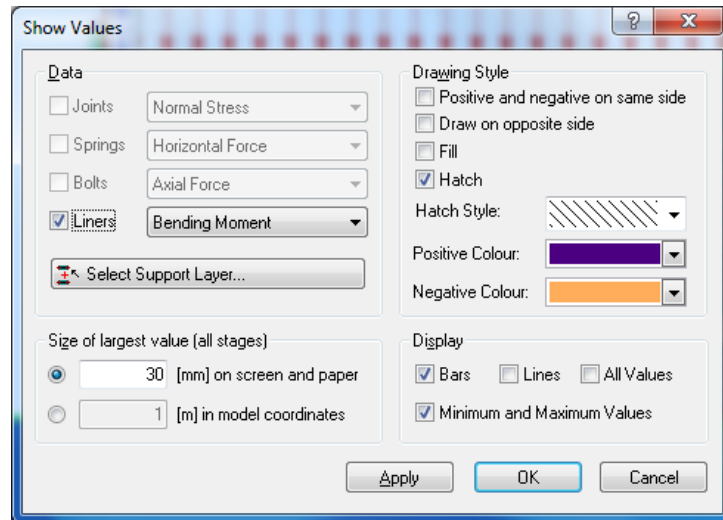
Note the buckling of the structure as the middle displaces downwards and the edges bulge outwards.



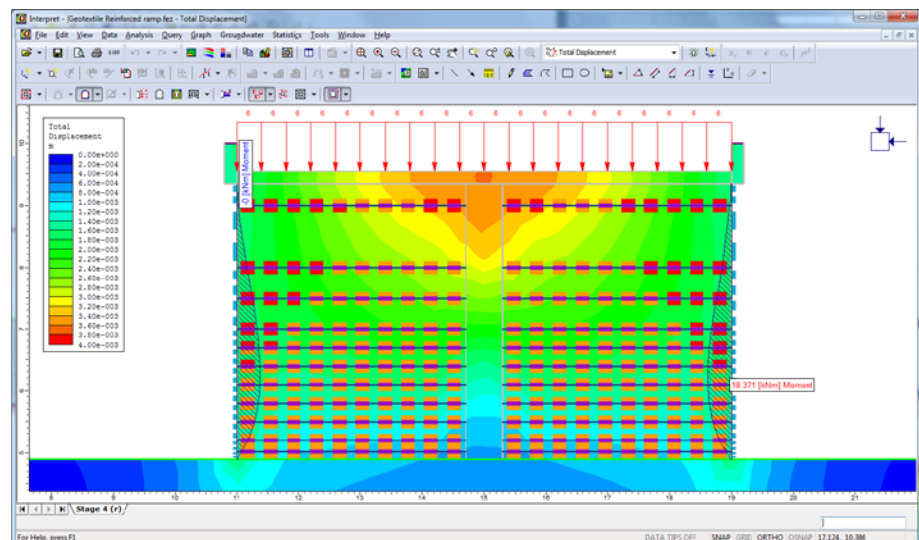
Check if any of the supports have failed by clicking on the Display Yielded Liners button. You will see that the outside edges of the upper geotextiles have failed. The precast concrete liners have remained intact since we set them to be elastic.

NOTE: If you want to know the manner in which the geotextile liners have failed then turn on the Expand Composites option in the Display Options dialog. You will see that the liner failure is in fact slipping of the joints of the structural interface. The actual geotextile component of the structural interface has not failed (i.e. its tensile strength has not been exceeded).

You can easily plot the bending moments associated with the concrete supports. First turn off the deformed boundaries. Now go to Show Values in the Analysis menu and select Show Values. Beside the Liners check box, choose Bending Moment from the drop down menu.



Click OK and you will see the bending moments plotted graphically on the model.



The maximum and minimum moments are shown. Note that the geotextiles cannot support moments so all of their values are 0. If you want to see the true minimum moment for the concrete supports, then turn off the plots for the geotextiles by going to Show Values from the Analysis menu and choosing Toggle Show Values. Now click on each of the geotextiles to turn them off.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.

Additional Exercise

Try setting the Material Type of the Foundation Soil to Plastic and see if there is any change in the results.